Advanced Optical Fiber Communications

Problem 1: Wavelength

A plane electromagnetic wave oscillates at an angular frequency of $1.216 \cdot 10^{15}$ rad/s.
Find its wavelength in free space and in the following media:
Water ($n_{H2O} = 1.33$), glass (SiO_2, $n_{SiO2} = 1.44$) and Gallium Arsenide (GaAs, $n_{GaAs} = 3.6$).

Problem 2: Wave equation

a) Show that $\underline{\vec{E}}(z,t) = \begin{pmatrix} E_x e^{j(\omega t - \underline{k}z)} \\ 0 \\ 0 \end{pmatrix}$ is a solution of the wave equation $\nabla^2 \underline{\vec{E}} + \mu_0 \varepsilon_0 \underline{\varepsilon}_r \omega^2 \underline{\vec{E}} = 0$
and determine the expression for $\underline{k}$.

b) Use $\underline{k} = \beta - j\alpha$ and determine the real part of $\underline{E}(z,t)$.
Sketch $\mathrm{Re}\{\underline{E}(z,t)\}$ for $\omega t = 0$ and $\omega t = \pi/2$.

c) Use the characteristic impedance $\underline{Z}$ to calculate the magnetic field. What does a complex $\underline{Z}$ mean for the phase relation between $\underline{\vec{E}}(z,t)$ and $\underline{\vec{H}}(z,t)$?

Problem 3: Polarization

Consider the electrical field of a plane wave with two components:

$$\vec{E}(z,t) = \begin{pmatrix} E_x \cos(\omega t - kz) \\ E_y \cos(\omega t - kz + \varphi) \\ 0 \end{pmatrix}$$

Sketch the vector in the xy-plane at z = 0 for $\varphi = 0$ and $\varphi = \pi/2$ for different times.
Show that for $\varphi = \pi/2$ and $E_x = E_y$ the vector rotates on a circle.